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ESTIMATING THE LIGHTING OF A HOUSE

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Introduction

In modern engineering, accurate planning is essential for designing efficient electrical systems, especially for building lighting. This project estimates the lighting requirements of a residential building using AutoCAD, a widely used software in architectural and electrical design. Electrical symbols are placed on a 2D floor plan according to standard illuminance levels in order to determine the number and type of luminaires for each room.

The building area, ceiling height and room layout are taken into account to achieve proper light distribution, using the lumen method. The project then groups the luminaires into lighting circuits, selects the cables and protective devices, and finishes with a bill of quantities that estimates the materials, labour and total installation cost.

Overall, the project provides a clear method for planning and estimating an effective and economical lighting system for a house.

Design data and assumptions

Parameter	Value
Ceiling height	3.0 m
Work-plane height	0.8 m → mounting height H_m = 3.0 − 0.8 = 2.2 m
Lamp	20 W, 2500 lm (F = 2500 lm)
Maintenance factor (MF)	0.8
Reflectances	Ceiling C = 0.7, Wall W = 0.5, Floor F = 0.3
Supply	230 V, 50 Hz, single phase (power factor taken as 1 for the lamps)

Tables that I used for my calculations

Table 1: Standard range of lux levels

Room	Recommended illuminance (E)
Living Room	150–300 lx
Kitchen	200–400 lx
Dining Room	150–250 lx
Bedroom	150–200 lx
Bathroom	200–300 lx
Hallway	100–150 lx
Staircase	150 lx
Study / Office	400–500 lx
Storage Room	100–150 lx

Values adopted in this project: bedrooms 150 lx, bathrooms / dressing room / entrance 200 lx, living room with open kitchen 300 lx. The sport room is not listed in Table 1, so 400 lx (the Study / Office value) was assumed for it.

Table 2: Utilization factor

Reflectance			Room Index								
C	W	F	0.75	1.00	1.25	1.50	2.00	2.50	3.00	4.00	5.00
0.7	0.5	0.0	0.64	0.68	0.71	0.74	0.77	0.79	0.80	0.83	0.85
		0.1	0.66	0.71	0.74	0.77	0.80	0.83	0.85	0.88	0.90
		0.2	0.67	0.73	0.77	0.80	0.84	0.87	0.90	0.93	0.96
		0.3	0.69	0.75	0.80	0.84	0.89	0.92	0.95	0.98	0.99
	0.3	0.0	0.59	0.63	0.66	0.68	0.72	0.74	0.76	0.79	0.81
		0.1	0.60	0.64	0.68	0.71	0.75	0.77	0.79	0.83	0.85
		0.2	0.61	0.66	0.70	0.73	0.78	0.81	0.83	0.87	0.91
		0.3	0.62	0.68	0.72	0.75	0.81	0.85	0.88	0.93	0.96
	0.1	0.0	0.55	0.59	0.62	0.64	0.68	0.70	0.72	0.75	0.77
		0.1	0.56	0.60	0.63	0.66	0.70	0.72	0.75	0.78	0.81
		0.2	0.56	0.61	0.64	0.67	0.72	0.75	0.78	0.82	0.86
		0.3	0.57	0.62	0.66	0.69	0.74	0.78	0.82	0.87	0.91

Row used for this project (C = 0.7, W = 0.5, F = 0.3):

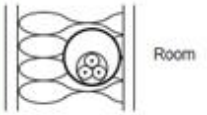
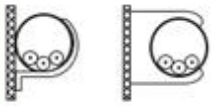
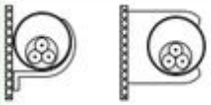
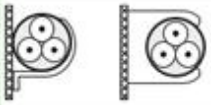
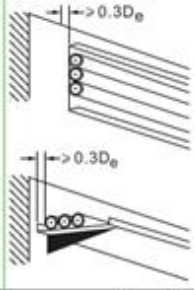
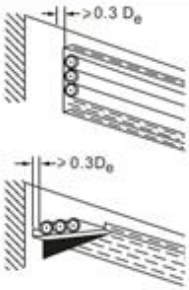
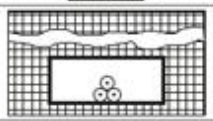
Room index k	0.75	1.0	1.25	1.5	2.0
UF	0.69	0.75	0.80	0.84	0.89

For values of k between two columns the utilization factor is found by linear interpolation. When k is lower than 0.75 (the first column), the lowest tabulated value 0.69 is used.

Table 3: Current-carrying capacity in amperes

Reference methods	Number of loaded conductors and type of insulation											
A1		2 PVC	3 PVC		3 XLPE	2 XLPE						
A2	3 PVC	2 PVC		3 XLPE	2 XLPE							
B1				3 PVC	2 PVC		3 XLPE		2 XLPE			
B2			3 PVC	2 PVC		3 XLPE	2 XLPE					
C					3 PVC		2 PVC	3 XLPE		2 XLPE		
E						3 PVC		2 PVC	3 XLPE		2 XLPE	
F							3 PVC		2 PVC	3 XLPE		2 XLPE
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
Size (mm^2) Cooper												
1.5	13	13.5	14.5	15.5	17	18.5	19.5	22	23	24	26	-
2.5	17.5	18	19.5	21	23	25	27	30	31	33	36	-
4	23	24	26	28	31	34	36	40	42	45	49	-
6	29	31	34	36	40	43	46	51	54	58	63	-
10	39	42	46	50	54	60	63	70	75	80	86	-
16	52	56	61	68	73	80	85	94	100	107	115	-
25	68	73	80	89	95	101	110	119	127	135	149	161
35	-	-	-	110	117	126	137	147	158	169	185	200
50	-	-	-	134	141	153	167	179	192	207	225	242
70	-	-	-	171	179	196	213	229	246	268	289	310
95	-	-	-	207	216	238	258	278	298	328	352	377
120	-	-	-	239	249	276	299	322	346	382	410	437
150	-	-	-	-	285	318	344	371	395	441	473	504
185	-	-	-	-	324	362	392	424	450	506	542	575
240	-	-	-	-	380	424	461	500	538	599	641	679
Aluminium												
2.5	13.5	14	15	16.5	18.5	19.5	21	23	24	26	28	-
4	17.5	18.5	20	22	25	26	28	31	32	35	38	-
6	23	24	26	28	32	33	36	39	42	45	49	-
10	31	32	36	39	44	46	49	54	58	62	67	-
16	41	43	48	53	58	61	66	73	77	84	91	-
25	53	57	63	70	73	78	83	90	97	101	108	121
35	-	-	-	86	90	96	103	112	120	126	135	150
50	-	-	-	104	110	117	125	136	146	154	164	184
70	-	-	-	133	140	150	160	174	187	198	211	237
95	-	-	-	161	170	183	195	211	227	241	257	289
120	-	-	-	186	197	212	226	245	263	280	300	337
150	-	-	-	-	226	245	261	283	304	324	346	389
185	-	-	-	-	256	280	298	323	347	371	397	447
240	-	-	-	-	300	330	352	382	409	439	470	530

Table 4: Examples of methods of installation

Item No	Method of installation	Description	Reference method of installation to be used to obtain current-carrying capacity
1		Insulated conductors or single-core cables in conduit in a thermally insulated wall	A1
2		Multi-core cables in conduit in a thermally insulated wall	A2
4		Insulated conductors or single-core cables in conduit on a wooden, or masonry wall or spaced less than 0.3 x conduit diameter from it	B1
5		Multi-core cable in conduit on a wooden, or mansonry wall or spaced less than 0.3 x conduit diameter from it	B2
20		Single-core or multi-core cables: - fixed on, or sapced less than 0.3 x cable diameter from a wooden wall	C
30		Single-core or multi-core cables: on unperforated tray run horizontally or vertically	C
31		Single-core or multi-core cables: on perforated tray run horizontally or vertically	E or F
36		Bare or insulated conductors on insulators	G
70		Multi-core cables in conduit or in cable ducting in the ground	D1
71		Single-core cable in conduit or in cable ducting in the ground	D1

Equations that I used in my calculations

Equation 1: Room index

$$k = \frac{L \times W}{H_m (L + W)}$$

k: room index **L**: room length (m) **W**: room width (m) $\mathbf{H_m}$: mounting height = ceiling height − 0.8 m

Equation 2: Number of lamps

$$n = \frac{A \times E}{F \times UF \times MF}$$

n: number of lamps required **A**: area of the room (m²) **E**: recommended illuminance (lx) **F**: luminous flux of one lamp (lm) **UF**: utilization factor **MF**: maintenance factor

Equation 3: Distance between two adjacent luminaires

$$S = \frac{L}{n_{row}}$$

S: spacing (m) **L**: room length (m) $\mathbf{n_{row}}$: number of luminaires in one row

Equation 4: Illuminance achieved with the adopted number of lamps

$$E_{achieved} = \frac{n \times F \times UF \times MF}{A}$$

Used to check that the adopted number of lamps meets the required illuminance.

Equation 5: Load current of a circuit

$$I = \frac{P}{V}$$

I: current (A) **P**: power (W) **V**: supply voltage = 230 V

Equation 6: Voltage drop (single-phase)

$$\Delta V = \frac{2 \rho L I}{A_c}$$

ρ: resistivity of copper (0.0225 Ω·mm²/m at 70 °C) **L**: one-way cable length (m) **I**: current (A) $\mathbf{A_c}$: cable cross-section (mm²)

Calculations

The same procedure is applied to every room: (1) find the room index k, (2) read the utilization factor from Table 2 for C = 0.7, W = 0.5, F = 0.3, (3) calculate the number of lamps n, round it to a practical whole number, and (4) check the illuminance actually achieved. Constants for all rooms: F = 2500 lm, MF = 0.8, H_m = 3 − 0.8 = 2.2 m.

Bedroom 1 = Bedroom 2 = Bedroom 3 (4.5 × 3.8 × 3 m)

A = 4.5 × 3.8 = **17.10 m²** E = **150 lx** (Table 1)

k = (4.5 × 3.8) / [2.2 × (4.5 + 3.8)] = 17.10 / 18.26 = **0.94**

UF = **0.73** (interpolated from Table 2)

n = (17.10 × 150) / (2500 × 0.73 × 0.8) = **1.76**

Adopted: **2 lamps** (achieved E = 171 lx ≥ 150 lx).

Distance between adjacent luminaires S = 4.5 / 2 = **2.25 m**

Bathroom 1 = Bathroom 2 = Dressing Room = Entrance (2.9 × 2.0 × 3 m)

A = 2.9 × 2.0 = **5.80 m²** E = **200 lx** (Table 1)

k = (2.9 × 2.0) / [2.2 × (2.9 + 2.0)] = 5.80 / 10.78 = **0.54**

UF = **0.69** (k is below 0.75, so the lowest value in Table 2 is used)

n = (5.80 × 200) / (2500 × 0.69 × 0.8) = **0.84**

Adopted: **1 lamp** (achieved E = 238 lx ≥ 200 lx).

Distance between adjacent luminaires S = 2.9 / 1 = **2.9 m**

Note: the dressing room measures 2.35 × 2.9 m on the plan; it gives k = 0.59, UF = 0.69 and n = 0.99, so one lamp is still adequate.

Sport Room (4.5 × 2.3 × 3 m)

A = 4.5 × 2.3 = **10.35 m²** E = **400 lx** (Table 1)

k = (4.5 × 2.3) / [2.2 × (4.5 + 2.3)] = 10.35 / 14.96 = **0.69**

UF = **0.69** (k is below 0.75, so the lowest value in Table 2 is used)

n = (10.35 × 400) / (2500 × 0.69 × 0.8) = **3.00**

n = 3.0 is the minimum; **4 lamps** are adopted for a symmetrical 2 × 2 layout (achieved E = 533 lx).

Distance between adjacent luminaires S = 4.5 / 2 = **2.25 m**

Living Room + Open Kitchen (9.6 × 5.3 × 3 m)

A = 9.6 × 5.3 = **50.88 m²** E = **300 lx** (Table 1)

k = (9.6 × 5.3) / [2.2 × (9.6 + 5.3)] = 50.88 / 32.78 = **1.55**

UF = **0.85** (interpolated from Table 2)

n = (50.88 × 300) / (2500 × 0.85 × 0.8) = **8.98**

n = 8.98 ≈ **9 lamps** (achieved E = 301 lx ≥ 300 lx).

Distance between adjacent luminaires S = 9.6 / 4 = **2.4 m**

Summary of lighting load

Location	A (m²)	E (lx)	k	UF	n calc.	n adopted	Power (W)	Current (A)	Lamp cost ($)
Bedroom 1	17.10	150	0.94	0.73	1.76	2	40	0.17	4.00
Bedroom 2	17.10	150	0.94	0.73	1.76	2	40	0.17	4.00
Bedroom 3	17.10	150	0.94	0.73	1.76	2	40	0.17	4.00
Bathroom 1	5.80	200	0.54	0.69	0.84	1	20	0.09	2.00
Bathroom 2	5.80	200	0.54	0.69	0.84	1	20	0.09	2.00
Dressing Room	5.80	200	0.54	0.69	0.84	1	20	0.09	2.00
Entrance	5.80	200	0.54	0.69	0.84	1	20	0.09	2.00
Sport Room	10.35	400	0.69	0.69	3.00	4	80	0.35	8.00
Living Room with Open Kitchen	50.88	300	1.55	0.85	8.98	9	180	0.78	18.00
Total						**23**	**460**	**2.00**	**46.00**

Current is calculated with I = P / V and V = 230 V (e.g. 40 W / 230 V = 0.17 A). Total: 460 W / 230 V = 2.00 A. Lamp cost is $2 per 20 W lamp (23 lamps = $46).

Procedure

1. First, I designed my house map.

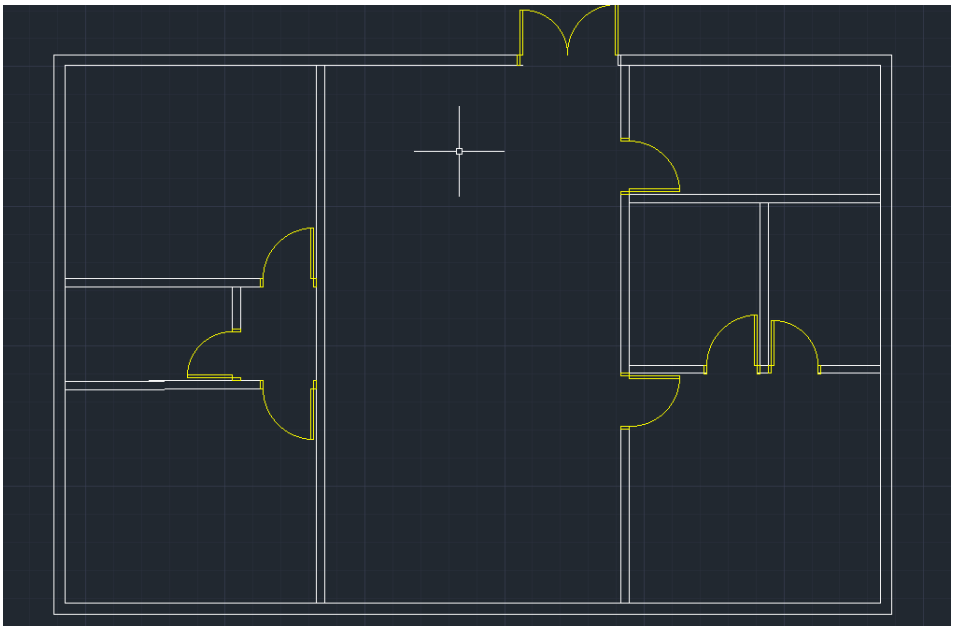


Figure 1: House plan

2. Then, I added the dimensions and the name of each room.

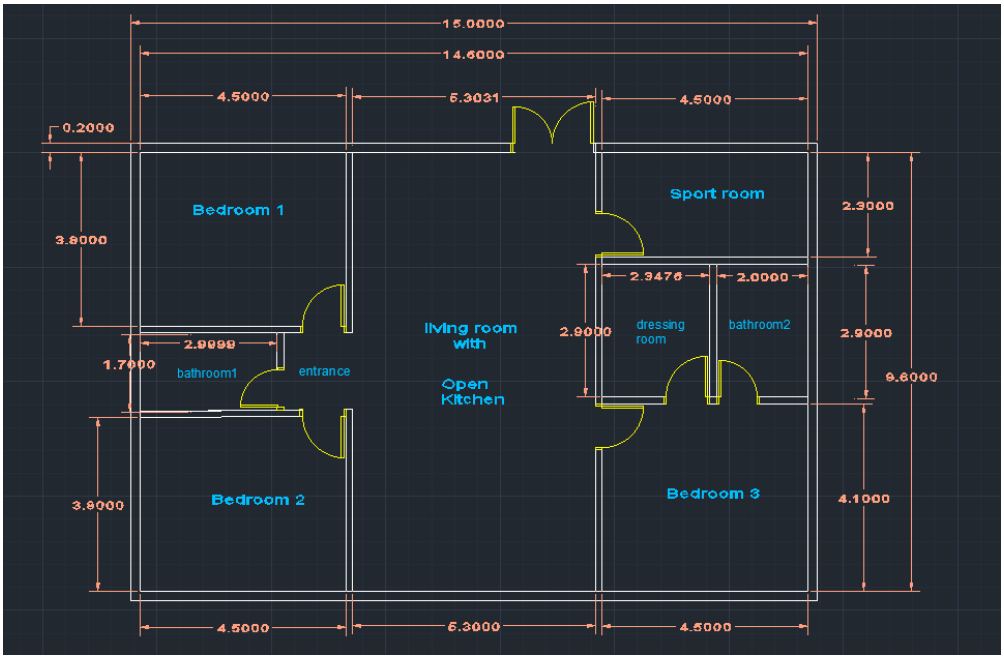


Figure 2: Plan with dimensions and room names

3. Then, I added the lights and the switches of each room.

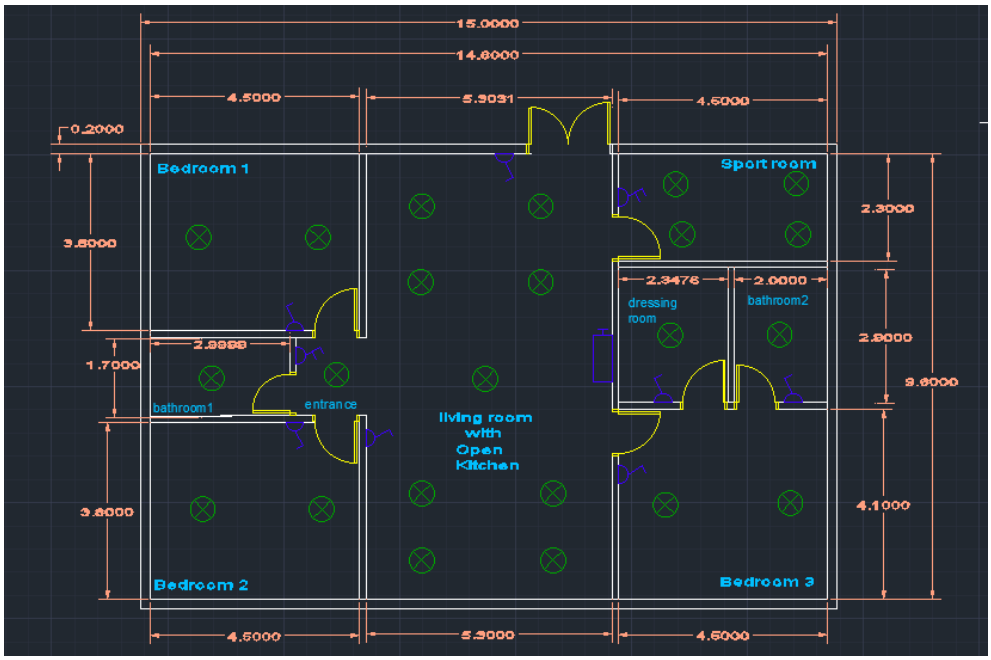


Figure 3: Luminaires and switches

4. Electrical connection between the sub-main switch and the single and double (two-gang) one-way switches.

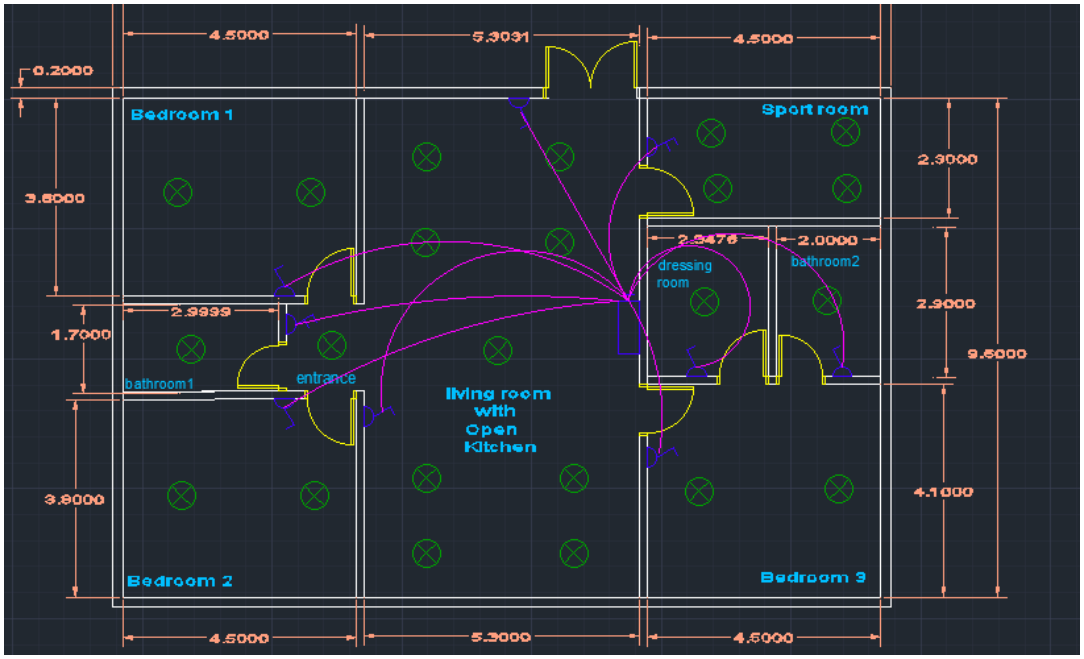


Figure 4: Connection of the sub-main switch to the switches

5. Finally, my electrical design of the house lighting.

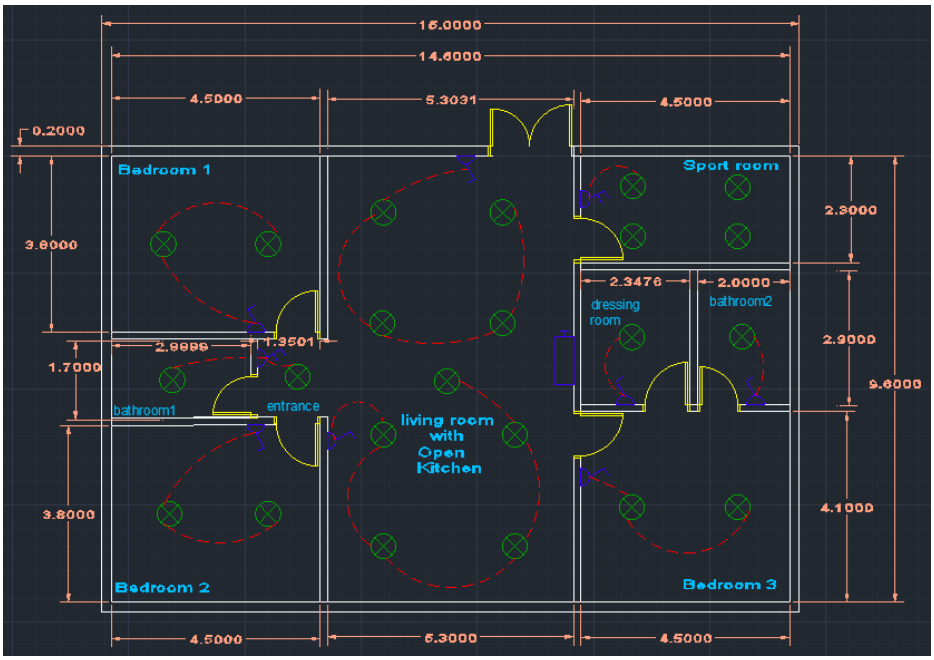


Figure 5: Final lighting design with dimensions

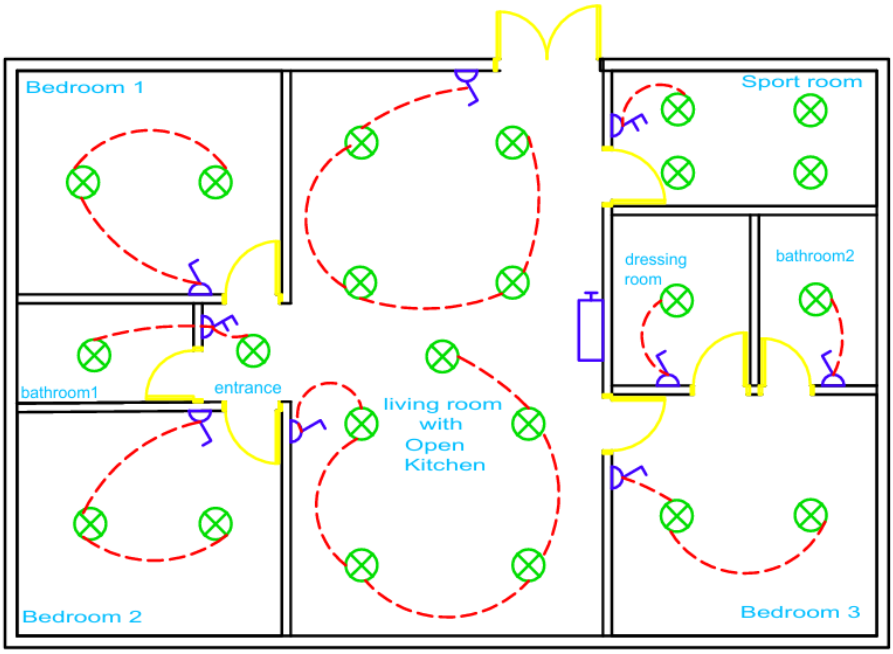


Figure 6: Final lighting design (clean view)

Table of graphical symbols

Symbol	Description
	**20 W lamp**
	**Single-gang one-way switch**
	**Double-gang one-way switch**
	**Sub-main switch (distribution board)**

Lighting circuits and cable selection

Lighting circuits

The 23 lamps are supplied from a single distribution board (DB) and divided into three lighting circuits, so that a fault in one area does not switch off the whole house. Each circuit is protected by a 6 A miniature circuit breaker (MCB).

Circuit	Area supplied	Lamps	Power (W)	Current (A)	MCB
L1	Bedrooms 1, 2 and 3	6	120	0.52	6 A
L2	Bathrooms 1 and 2, dressing room, entrance, sport room	8	160	0.70	6 A
L3	Living room with open kitchen	9	180	0.78	6 A
Total		**23**	**460**	**2.00**	

Cable selection

From Table 3, a **1.5 mm² copper PVC cable** is chosen for all lighting circuits. The highest circuit current is 0.78 A (circuit L3) and the total current is 2.00 A, while the capacity of a 1.5 mm² copper cable in Table 3 is at least 13 A. The design condition $I_b \leq I_n \leq I_z$ is satisfied for every circuit (0.78 A ≤ 6 A ≤ 13 A). Lighting circuits are supplied from the DB to the junction boxes, and from each junction box to the lamps and switches of the room, so the installation methods A1 (single-core cables in conduit) and A2 (multi-core cables in conduit) of Table 4 are used.

Path	Material	Insulation	Cores	Cross-section (mm²)
DB to junction boxes	Copper	PVC	Single core (A1)	1.5
Junction boxes to lamps and switches	Copper	PVC	Multi core (A2)	1.5

Copper: normally used for internal wiring in houses; long life and high reliability. **PVC:** sufficient insulation for indoor conduit installation.

Voltage-drop check

For the most loaded circuit (L3, I = 0.78 A) and an assumed longest run of 25 m:

$$\Delta V = (2 \times 0.0225 \times 25 \times 0.78) / 1.5 = \mathbf{0.58\ V} = 0.25\ \%\ \text{of}\ 230\ \text{V}$$

This is far below the 3 % limit usually applied to lighting circuits, so 1.5 mm² is adequate.

Cost estimation

The bill of quantities below estimates the materials, labour and total installation cost of the lighting system. Cable and conduit lengths are estimated from the plan (about ±10 %, including 10 % waste): L1 ≈ 45 m, L2 ≈ 55 m and L3 ≈ 45 m of cable, giving about 145 m in total.

No.	Item	Unit	Qty	Unit price ($)*	Amount ($)
1	20 W lamp, 2500 lm	pcs	23	2.00	46.00
2	One-way switch (single / double gang)	pcs	9	1.20	10.80
3	Junction box	pcs	9	0.50	4.50
4	Copper PVC cable 1.5 mm²	m	145	0.35	50.75
5	PVC conduit 20 mm	m	120	0.30	36.00
6	Distribution board, 6-way, with main switch / RCD	pcs	1	25.00	25.00
7	MCB 6 A	pcs	3	3.00	9.00
8	Accessories (clips, saddles, tape)	lot	1	10.00	10.00
	Materials subtotal				**192.05**
	Labour (35 % of materials)				**67.22**
	Subtotal				**259.27**
	Overhead and profit (10 %)				**25.93**
	TOTAL ESTIMATED COST				**285.19**

* Unit prices are assumed local market prices in US dollars; the lamp price ($2) is the one used in the first version of this project. The estimated total cost of the lighting installation is **about $285**.

Conclusion

In this project the lighting of a nine-room house was designed and estimated. Using the lumen method, 23 lamps of 20 W (2500 lm) were required, giving a total connected load of 460 W and a total current of 2.00 A at 230 V. The lamp layout, switches and wiring were drawn in AutoCAD, and the lamps were divided into three lighting circuits, each protected by a 6 A MCB and wired with 1.5 mm^2 copper PVC cable. The voltage drop (≈ 0.25 %) is far below the permitted limit.

The bill of quantities gives an estimated installation cost of about $285, including materials, labour, overhead and profit. Such an estimate shows the needs of the electrical system before installation, which allows better planning and coordination, and helps to detect possible clashes with other services such as water and drainage so that adjustments can be made in advance.

Accurate cost estimation is a critical part of any construction project because it directly affects its economic feasibility and overall success.